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The Effect of Visualization on Mental Models of Polarization

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Overview

1. Mental models and model understanding
2. Context: Polarization and the Poincaré sphere
3. Hypotheses and study design
4. Analysis and current status

Mental models and model understanding

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Gestalt and Functionality

Gestalt

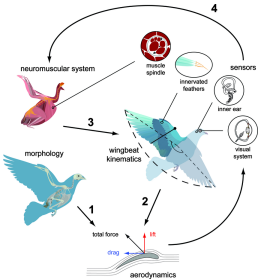
The set of forms and structures that a learner's mental model consists of.

Functionality

The set of functions that a learner's mental model consists of.

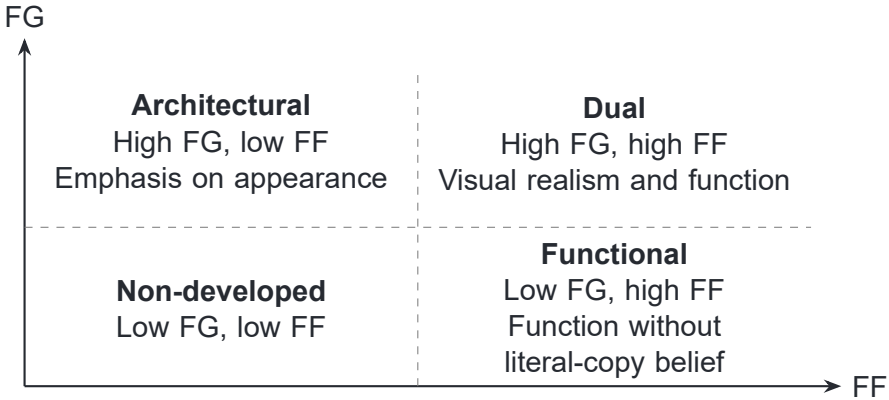


(WikimediaCommons, 2024)



(Altshuler et al., 2015)

Four aspects of model understanding



(Ubben and Bitzenbauer, 2022).

Fidelity and alignment answer different questions

Fidelity: the learner's stance

- What does the learner believe the model is faithful to?
- FG: appearance; FF: function.

Alignment: expert correspondence

- **Structural alignment (StA):** correspondence with expert-selected forms and structures.
- **Functional alignment (FA):** correspondence with expert-selected functions.

Visualization and the multimedia effect

Multimedia principle

Combining words with relevant pictures can benefit learners more than words alone.

(Mayer, 2005)

DeFT: Design, Functions, Tasks

- **Design:** The number of representations, how information is distributed, the form, the order, and how the translations are highlighted.
- **Functions:** Representations can **complement** each other, **constrain** interpretations, and support the construction of **deeper understanding**.
- **Tasks:** The cognitive tasks required for learning with multiple representations: understanding the representations, their relationships, and selecting accordingly.

(Ainsworth, 2006)

Two pathways for an additional visualization

Representational support

- Complement by allowing different strategies to solve problems.
- Connect notation, geometry, and optical transformations.
- Support expert-aligned explanations and predictions.

Spontaneous reflection

- Encounter different representations of the same state.
- Reconsider literal interpretations.
- Attend to shared functions and limits.

Due to the multimedia principle, expert alignment may improve even when fidelities remain stable.

Context: Polarization and the Poincaré sphere

Jones vectors and the Poincaré sphere

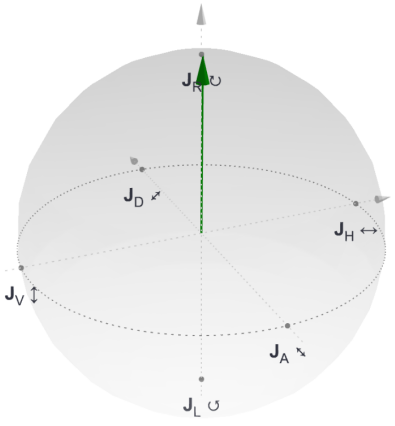
For a normalized, fully polarized state,

$$\mathbf{J} = \begin{pmatrix} a \\ b\, e^{i\varphi} \end{pmatrix}, \qquad a, b \geq 0, \quad a^2 + b^2 = 1.$$

The applet uses

$$\mathbf{S} = \begin{pmatrix} a^2 - b^2 \\ 2ab\cos\varphi \\ 2ab\sin\varphi \end{pmatrix}, \qquad \|\mathbf{S}\| = 1,$$

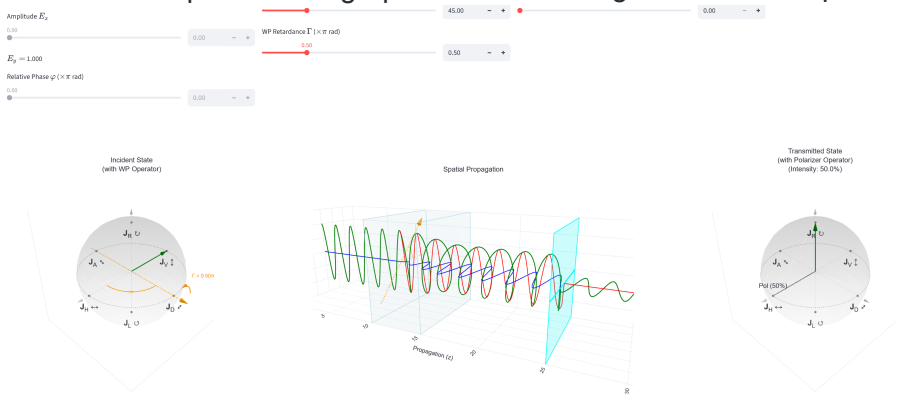
akin to a Qubit on the Bloch sphere, a valid approach to introductory quantum computing (Tóth et al., 2025) $|\psi\rangle = \cos\frac{\vartheta}{2}|0\rangle + e^{i\varphi}\sin\frac{\vartheta}{2}|1\rangle$



The applet

Control:
Jones notation and spatial wave graph.

Experimental:
Adding the Poincaré sphere.



+ Introductory explanations, guided parameter changes, hints, and solutions.

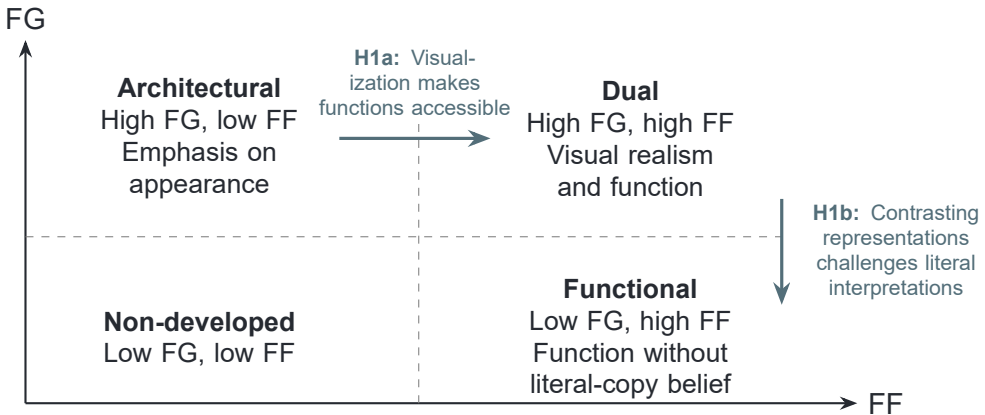
The learning sequence

Steps	Topic	Learner activity
1–3	Polarization states	Change amplitudes and relative phase to construct linear and circular states.
4–6	Wave plates	Vary retardance and fast-axis orientation to transform a state.
7	Linear polarizers	Relate transmission-axis orientation to output intensity.
8–9	Combined operators	Coordinate wave plates and polarizers to meet a target.

Nine challenges precede a completion screen and survey return link.

Hypotheses and study design

Possible changes in model understanding



Research questions

Research Question 1

(How) does the Poincaré sphere visualization change thinking in Fidelity of Gestalt (FG) and/or Functional Fidelity (FF) internal models on polarization?

Research Question 2

(To what extent) does the Poincaré sphere visualization increase alignment with expert views on Gestalt and Function?

Research Question 3

(To what extent) are the interactions mediated by cognitive load, spatial reasoning ability, and baseline FG?

Hypotheses

H1a: Including the architectural profile

For learners with high FG and low FF, the sphere may make functions accessible: Wave plates act as **rotations** of the state, while polarizers perform **projections** of the Jones state.

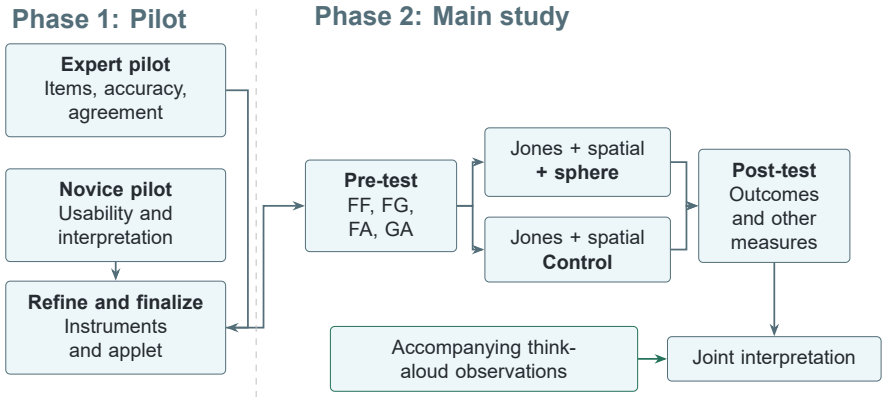
H1b: Including the dual profile

For learners with high FG and high FF, contrasting the spatial wave representation with the abstract sphere may challenge literal interpretations and reduce FG while FF stays high.

H2: Multimedia effect

The support with an additional visual representation may increase expert alignment (Rexigel et al., 2024).

Study structure



Phase 1: Calibrate the instruments and the applet

Expert pilot

- Classify items as FF, FG, both, or neither.
- Judge factual correctness and the usefulness of analogies.
- Evaluate wording and agreement.

Novice pilot

- Identify confusing controls and instructions.
- Examine interpretation of the representations.
- Assess cognitive load (Krieglstein et al., 2023) and technological acceptance (Abdullah and Ward, 2016).

Refine questionnaire items, applet instructions, and interaction design before the main comparison.

Phase 2: Randomized pre/post comparison

Pre-test	FF, FG, FA, and GA
Intervention	Random assignment to guided polarization tasks with or without the added sphere
Post-test	FF, FG, FA, GA; cognitive load (Krieglstein et al., 2023) and spatial reasoning (Young et al., 2025), demographic information

Population: second- and third-year physics / related STEM undergraduates.

Target: Power analysis: $N = 218$ recruited; minimum analyzable sample $N = 196$.
We need your help to recruit participants for the online study!

Measures and protocol decisions

- Cognitive load: Extraneous (5 questions), Intrinsic (5) and Germane (7), 9-point Likert scale (Krieglstein et al., 2023)
- General Extended Technology Acceptance Model for E-Learning (GETAMEL, only in pilot study, 7-point Likert scale): (Abdullah and Ward, 2016)
  - Perceived Usefulness
  - Perceived Ease of Use
  - E-Learning Self-Efficacy
  - Perceived Enjoyment
- Spatial reasoning: Measured using the open-source 3D Block Rotation scale from the International Cognitive Ability Resource (ICAR) (Young et al., 2025)

Think-aloud observations: Investigating mechanisms

Observe the students'

- attempts to incorporate the Poincaré sphere into their thinking.
- difficulties in connecting representations.
- Expressed conflicts.
- Conflict resolution.

The observations can provide evidence for mechanisms underlying learning and limtations of mental models.

Analysis and current status

Planned primary analysis: MANCOVA

Joint outcomes

$$\mathbf{Y}_{\text{post}} = \begin{pmatrix} FF_{\text{post}} \\ FG_{\text{post}} \\ StA_{\text{post}} \\ GA_{\text{post}} \end{pmatrix}$$

- **Factor:** condition, with or without the additional Poincaré sphere.
- **Covariates:** baseline FF, FG, StA, and GA, cognitive load, and spatial reasoning ability to assess RQ 2.

Technological Acceptance: Perceived Usefulness and Ease of Use

Perceived Usefulness

- Interactive graphics made the effects of parameter changes immediately visible.
- Hints and direct experimentation supported learning.
- Missing examples and definitions reduced the perceived learning value.

Perceived Ease of Use

- Camera resets, scrolling, and perspective controls disrupted interaction.
- Sliders were sometimes difficult to adjust precisely.
- Participants wanted easier access to instructions, formulas, and previous content.

Technological Acceptance: Self-Efficacy and Enjoyment

Self-Efficacy

- Hints enabled participants to proceed independently.
- Some completed the tasks despite uncertainty about matrices and terminology.
- Task completion did not necessarily indicate conceptual confidence.

Enjoyment

- Participants enjoyed manipulating parameters and observing graphical changes.
- Solving successive tasks created motivation similar to a reward system.
- Complex content and awkward controls reduced enjoyment.

Outlook

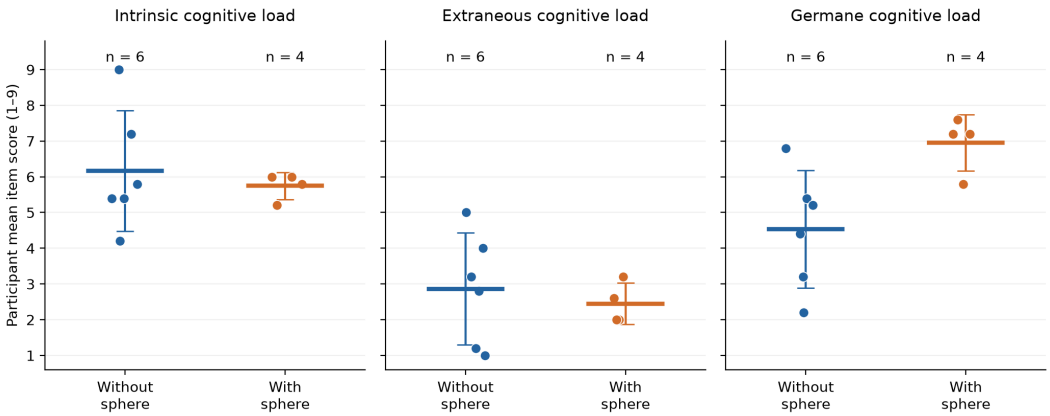
Pilot study takeaways:

- Interactivity and visualization were valued most.
- Usability problems and insufficient conceptual guidance were the main barriers.
- The comments do not reveal a distinct benefit of the Poincaré sphere.

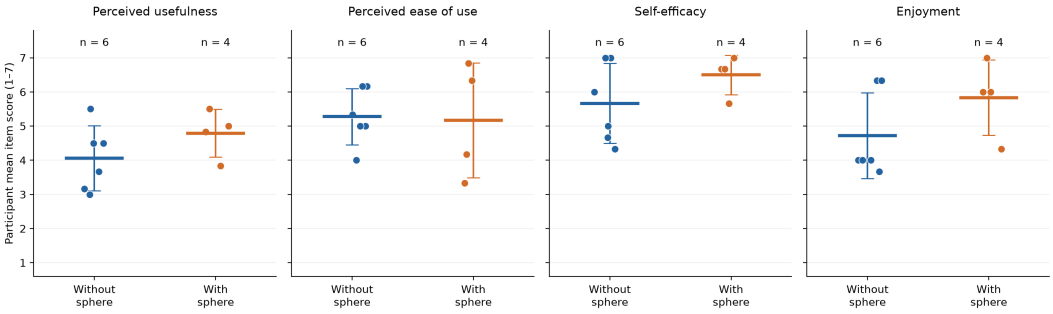
Next steps:

- Collect the remaining expert data and conduct the main study.
- Conduct the planned MANCOVA and follow-up tests.
- Integrate quantitative outcomes with think-aloud observations to investigate learning mechanisms.
- Report findings and implications for model understanding and the visualizations in physics education.

Initial pilot findings: cognitive load



Initial pilot findings: technological acceptance



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